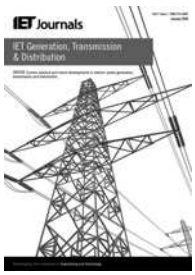
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Published in IET Generation, Transmission & Distribution
Received on 5th November 2012
Revised on 18th April 2013
Accepted on 19th April 2013
doi: 10.1049/iet-gtd.2012.0639



ISSN 1751-8687

A model for optimal scheduling of hydro thermal systems including pumped-storage and wind power

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Abstract: This study describes a model for optimal scheduling of hydro thermal systems with multiple hydro reservoirs. Inflow to hydropower reservoirs, wind power and exogenously given prices are treated as stochastic variables. Power flow constraints are included through a linearised power flow model. A linearised representation of start-up costs for generating units and pumps is provided. The model is well suited for medium- and long-term hydro thermal generation scheduling and has the capability of capturing detailed system constraints by using a fine time resolution. The presented model is tested on a realistic representation of the Icelandic power system, considering some potential future system extensions.

1 Introduction

There is a need for coupling efforts between traditional power system market modelling and hydro thermal scheduling on one side and transmission system modelling on the other. This need is frequently revealed in power markets through tighter market balances and increased power flow variations. An example is the large-scale introduction of wind power, which will often, because of the highly stochastic nature and possible remote locations, stress system constraints, such as transmission system and hydro reservoir capacities. In order to realistically capture this type of system constraints, traditional hydro thermal market models are challenged to cope with a finer time resolution.

A formal solution of the hydro thermal scheduling problem was presented some 50 years ago in [1, 2]. This approach is known as the incremental water value method and is based on stochastic dynamic programming (SDP). Owing to the dimensionality of the problem, SDP suffers from extreme computation times when considering systems with more than a few reservoirs. It is therefore most often applied to aggregated one-reservoir models of more complicated hydro systems and for simulation purposes supplemented by heuristics [3, 4].

Stochastic dual dynamic programming (SDDP) was introduced in [5, 6], and serves as an alternative to the SDP approach. In short, SDDP allows representation of many more state variables than SDP within reasonable computation times. It decomposes the multi-stage stochastic optimisation problem into one-stage sub-problems, where the expected future cost function is approximated by hyperplanes in the space of the state variables. The entire scheduling problem is then solved iteratively as an optimisation problem. Thus, more realistic and detailed models can be dealt with in a consistent manner than with the SDP-approach.

Applications of the SDDP algorithm on a detailed hydropower system while taking into account power flow constraints, have been documented in the literature [7, 8]. The work presented in [7] extends the algorithm in [6] to incorporate linearised power flow constraints. As pointed out in [7], the one-stage sub-problem can be seen as a linearised optimal power flow problem for the specified state and realisation of stochastic variables. A practical application of this algorithm is presented in [8], where emphasis is on expansion planning and identification of price zones. Later research related to the SDDP algorithm has, for example, focused on convergence properties [9], strategic bidding [10], risk aversion [11] and emission constraints [12]. The work presented in this paper primarily adds to recent research through the novel treatment of approximate start-up costs on pumps and generating units, and through the treatment of detailed wind power in the SDDP context.

The presented model comprises three extension to an earlier SDDP-based model for stochastic hydro thermal scheduling described in [13]. First, the modelling of linearised power flow constraints was added, as described in [7, 14]. Thus, power generation, demand and market purchases and sales need to be allocated individual busbars in a transmission network representation, and consequently power prices refer to each individual busbar, that is, as nodal prices. Details related to adopting the model to a nodal pricing scheme are discussed, both in terms of model building and result handling. Second, start-up costs for generating units and pumps are approximated by using a continuous variable for the operating state, as described in [15]. The operating state has to be entered into the cuts describing the future cost functions in the SDDP approach. The final extension regards the treatment of energy time-series of wind with a minimum time resolution of 1 h.

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Wind power stochasticity could then be taken into account in both the strategy computation and system simulation part of the SDDP algorithm. Consequently, the presented model is well suited for coping with finer time resolution in long- and medium-term hydro thermal generation scheduling, which is demonstrated in the case study.

This paper is organised as follows. In Section 2, the mathematical system model is elaborated. Section 3 outlines the solution approach, emphasising on the amendments and novel concepts introduced. The new model has been used for an analysis of the power system of Iceland, and this case study is presented in Section 4. Finally, Section 5 concludes this work.

2 System model

2.1 Basic system model

The objective of the scheduling is to maximise the expected value of income from power sales, minus generation and purchase costs. A typical scheduling horizon is three to five years, and a basic time step of 1 week is applied. This is suitable for describing inflow stochasticity, and the decomposition in the SDDP algorithm is therefore carried out on a weekly basis. However, because of the variation in load, wind speed, exogenous power price and so on, the week is further divided into K sub-periods. This division provides either aggregate or sequential sub-periods. Aggregate sub-periods are found by lumping together hours according to load levels within the week, whereas sequential sub-periods subdivides the week in chronological order. The values of the stochastic variables are considered given within the week.

For a given week t a vector $\boldsymbol{X}_t$ is defined, comprising all variables for that week, such as water releases, thermal generation, market purchases and sales and so on, except the vector of reservoir volumes $\boldsymbol{V}_t$. Associated with $\boldsymbol{X}_t$ there is a cost vector $\boldsymbol{C}_t$ comprising all direct costs for the week, such as cost of thermal generation and the prices for buying and selling power. When the model is used with an exogenously given price, that price will be a component in C_t; see Section 2.5. It is assumed that all costs and relationships are linear or piecewise linear. Income from power sales are considered negative cost. The overall objective is then to find an operating strategy to obtain

$$\min \mathbb{E}\left\{\sum_{t=1}^{\mathrm{T}} \boldsymbol{C}_t^{\mathrm{T}}\boldsymbol{X}_t - \Phi\left(\boldsymbol{V}_T\right)\right\} \tag{1}$$

The expectation is to be taken over the stochastic variables inflow, wind energy and exogenous price. The function $\Phi(\boldsymbol{V}_T)$ estimates the value of water left in the reservoirs at time T, the end of the study period. Since water left in a reservoir at the end of a week is carried over to the next week, the weekly water balances for the reservoirs become coupled in time, making the optimisation problem a dynamic one. Thus, the problem in (1) is a multi-stage stochastic optimisation problem, which may be efficiently solved by decomposition techniques [16]. The presented model uses the SDDP approach [6], which is a sampling-based variant of multi-stage Benders decomposition. An outer approximation of a convex future cost function is constructed for each time-stage. The decomposition technique will be outlined in Section 3.1.

For a given realisation of stochastic variables, the decomposition in the SDDP algorithm leads to a linear sub-problem for week t, described by an objective function (2) and constraints (3)–(8)

$$J_t = \min\left(\alpha_t + \boldsymbol{C}_t^{\mathrm{T}}\boldsymbol{X}_t\right) \tag{2}$$

The future expected cost function α_t is constrained by hyperplanes or cuts of type (6), and the construction of these will be elaborated in Section 3.3. In line with, for example [13, 17], we present the basic constraints in condensed form below

$$\boldsymbol{V}_t - \boldsymbol{A}_V\boldsymbol{X}_t - \boldsymbol{Q}_t\boldsymbol{z}_t = \boldsymbol{V}_{t-1} + \boldsymbol{m}_t \tag{3}$$

$$\boldsymbol{z}_t = \boldsymbol{\phi}\boldsymbol{z}_{t-1} + \boldsymbol{v}_t \tag{4}$$

$$\boldsymbol{B}_t\boldsymbol{X}_t = \boldsymbol{P}_F \tag{5}$$

$$\alpha_t + \left(\boldsymbol{\kappa}_t^r\right)^{\mathrm{T}}\boldsymbol{V}_t + \left(\boldsymbol{\mu}_t^r\right)^{\mathrm{T}}\boldsymbol{z}_t \geq b_t^r, \quad r = 1, \ldots, R \tag{6}$$

With the limits

$$\boldsymbol{X}_t^{\min} \leq \boldsymbol{X}_t \leq \boldsymbol{X}_t^{\max} \tag{7}$$

$$\boldsymbol{V}_t^{\min} \leq \boldsymbol{V}_t \leq \boldsymbol{V}_t^{\max} \tag{8}$$

where T is the number of weeks in study period; $\Phi(\boldsymbol{V}_T)$ is the function giving the value of water at the end of the planning horizon, in €; $\boldsymbol{V}_t$ is the vector of reservoir volumes at the end of week t, in Mm^3; $\boldsymbol{A}_V$ is the matrix given by hydrosystem topology (incidence matrix); $\boldsymbol{Q}_t$ is the diagonal matrix containing inflow standard deviations for week t, in Mm^3; $\boldsymbol{z}_t$ is the normalised inflow vector; $\boldsymbol{m}_t$ is the vector of mean inflow week t, in Mm^3; $\boldsymbol{\phi}$ is the inflow model transition matrix; $\boldsymbol{v}_t$ is inflow model noise vector; $\boldsymbol{B}_t$ is the matrix for power balances; $\boldsymbol{P}_F$ is the firm power demand, in MWh/h; α_t is the expected future cost function, at the end of week t; $\boldsymbol{\kappa}_t^r$ is the hydro storage cut coefficients for cut r, in €/Mm^3; $\boldsymbol{\mu}_t^r$ is the inflow cut coefficients for cut r, in €/Mm^3; b_t^r is the right-hand side constant for cut r, in €; R is the number of cuts in future cost function approximation at the end of week t.

The inflow $\boldsymbol{q}_t$ is given by

$$\boldsymbol{q}_t = \boldsymbol{Q}_t\boldsymbol{z}_t + \boldsymbol{m}_t \tag{9}$$

It is assumed that $\boldsymbol{z}_t$ follows the lag-1 vector autoregressive model (4). The inflow 'noise' vector $\boldsymbol{v}_t$ is assumed to have N_V discrete values, with a given probability for each. Note that this inflow model may generate negative inflows which should not be modified because of the convexity requirement of the SDDP algorithm. We have introduced artificial variables at high cost in the reservoir balances to compensate for this. For further discussions on the treatment of negative inflows in SDDP-based models, see [18, 19].

Equation (3) contains the water balances for the reservoirs after (9) has been substituted for the inflow. The matrix $\boldsymbol{A}_V$ is derived from the topology of the hydrosystem; it contains zeros in the columns not associated with releases or spills in $\boldsymbol{X}_t$. These equations are split into balances for each sub-period, so that $\boldsymbol{A}_V$ is composed of K blocks.

The system's power balances are of the form (5) and will be detailed in the next section. In Sections 2.3 and 2.4 new constraints for power transfer and for modelling start-up

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costs are introduced. Note that for notational convenience subscript t is generally omitted in the remainder of this section.

2.2 Power balance

The power balance in (5) is specified for each sub-period k in (10)

$$\boldsymbol{A}^k u^k - \boldsymbol{y}^{k+} + \boldsymbol{y}^{k-} + \boldsymbol{g}^k = \boldsymbol{P}_F^k - \boldsymbol{P}_W^k \tag{10}$$

Here $\boldsymbol{u}^k$, $\boldsymbol{y}^{k+}$, $\boldsymbol{y}^{k-}$ and $\boldsymbol{g}^k$ are sub-vectors of $\boldsymbol{X}_t$. All variables are in (or converted to) MWh/h. $\boldsymbol{u}^k$ denotes the releases in sub-period k, in Mm^3 The matrix $\boldsymbol{A}^k$ contains conversion factors for the hydropower stations, using piecewise linear descriptions. The $\boldsymbol{y}^k$ contains a series of options for selling or purchasing power, each with a given quantity and price. $\boldsymbol{y}^{k+}$ denotes sales in the power market and $\boldsymbol{y}^{k-}$ is purchases. $\boldsymbol{g}^k$ is a vector of thermal power generation. Wind power is a stochastic variable of the scheduling problem, and $\boldsymbol{P}_W^k$ is a realisation of this variable.

Some additional variables are needed to adopt to a nodal pricing scheme. $\boldsymbol{y}^k$ should comprise a curtailment option and a possibility to dump excess power at a negligible income at each busbar. The latter opportunity ensures that nodal prices are always positive, which again ensures that the piecewise linear approximation of the hydropower efficiency-curve is loaded in logical order.

2.3 Power flow constraints

A linearised (DC) power flow model is used to describe the power flow. This preserves convexity, which is required for the SDDP algorithm. The DC power flow algorithm is generally considered appropriate for technical-economical studies of the power system [20], and extensive testing indicates that its accuracy is relatively close to the AC power flow under a wide variety of operating conditions [21].

The flow constraints are implemented using power transfer distribution factors (PTDF), obtained from the power flow model. A PTDF for an individual busbar to a single line gives the ratio between the change of flow on the line and the change of power injection at the given busbar, provided that the change in injection is balanced by a predefined reference busbar.

Let the vector $\boldsymbol{\Theta}_\ell$ contain the PTDFs for line ℓ. If F_ℓ^k is the flow on the line (in MWh/h) and $\boldsymbol{P}_I^k$ a vector containing the net power injections at all busbars, then

$$F_\ell^k = \boldsymbol{\Theta}_\ell^{\mathrm{T}} \boldsymbol{P}_I^k \tag{11}$$

Now let $F_\ell^{k,\max}$ be the maximum flow limit on line ℓ and $\boldsymbol{X}^k$ a sub-vector of $\boldsymbol{X}_t$, both referring to sub-period k. For convenience a matrix $\boldsymbol{A}_I^k$ is defined so that

$$\boldsymbol{P}_I^k = \boldsymbol{A}_I^k \boldsymbol{X}^k - \boldsymbol{P}_F^k + \boldsymbol{P}_W^k \tag{12}$$

Using (12) the power flow constraint for line ℓ becomes

$$\begin{aligned} -F_\ell^{k,\max} + \boldsymbol{\Theta}_\ell^{\mathrm{T}}\left(\boldsymbol{P}_F^k - \boldsymbol{P}_W^k\right) \le \boldsymbol{\Theta}_\ell^{\mathrm{T}} \boldsymbol{A}_I^k \boldsymbol{X}^k \le F_\ell^{k,\max} \\ + \boldsymbol{\Theta}_\ell^{\mathrm{T}}\left(\boldsymbol{P}_F^k - \boldsymbol{P}_W^k\right) \end{aligned} \tag{13}$$

Note that quadratic line losses can be computed within the SDDP scheme, for example, by expressing the line losses as linear constraints, as elaborated in [22, 23]. Loss computation will not be further addressed here.

2.4 Start-up costs on generating units and pumps

The modelling of start-up costs on generating units and pumps is outlined below. Although emphasis is on pumps, the modelling of start-up costs on generating units follows the same structure.

Pumps are described with a simplified model, with a maximum and minimum pumping capacity. The costs incurred by starting pumps should ideally be modelled using binary variables. However, binary variables would conflict with the convexity requirement of the SDDP algorithm and would impose a massive increase in computation time. Thus, we adopt a linearised continuous representation of start-up costs as first suggested in [24], and later extended in [15]. A similar approach is documented in [25]. Note that the intention of introducing start-up costs in a long- and medium-term scheduling model is not to provide accurate commitment schedules, but rather to capture the general impact of these costs on system operation and price forecasts.

Utilisation of pumps in sub-period k in a given week is contained in the vector $\boldsymbol{u}^{P,k}$ which is a sub-vector of $\boldsymbol{u}^k$ in (10). For each pump j with a defined start-up cost and in each sub-period k the following three linear constraints are added to the sub-problem

$$P_j^{\min} u_{L,j}^{P,k} + \left(P_j^{\max} - P_j^{\min}\right) u_{H,j}^{P,k} - u_j^{P,k} = 0 \tag{14}$$

$$u_{L,j}^{P,k} - u_{H,k}^{P,k} \ge 0 \tag{15}$$

$$u_{L,j}^{P,k} - u_{L,j}^{P,k-1} - \delta_j^k \le 0 \tag{16}$$

$$0 \le u_{L,j}^{P,k} \le 1 \tag{17}$$

$$0 \le u_{H,j}^{P,k} \le 1 \tag{18}$$

where, for each unit j: $P_j^{\min}$ is the minimum pumping capacity, in MW; $P_j^{\max}$ is the maximum pumping capacity, in MW; $u_{L,j}^{P,k}$ is the per unit pumping below minimum capacity; $u_{H,j}^{P,k}$ is the per unit pumping above minimum capacity; δ_j^k is the start-up variable.

The coupling between the original pumping variable $u_j^{P,k}$ and the per unit variables $u_{L,j}^{P,k}$ and $u_{H,j}^{P,k}$ is given in (14). Equation (15) tries to load the pump correctly, ensuring that it is started before pumping above the minimum limit. Equation (16) defines the variable δ_j^k, which is used to assign a start-up cost on pumping. In practice, we relax (18) since there is already an upper limit on $u_j^{P,k}$.

Start-up costs are introduced into the overall model as a sum of contributions from all N_S pumps with start-up cost c_j^s. For 1 week this sum is

$$\sum_{k=1}^{K} \sum_{j=1}^{N_S} c_j^s \delta_j^k \tag{19}$$

Note that modelling start-up costs on a thermal generator will follow the same procedure as outlined above, that is, by adding three new variables and constraints of type (14)–(18) and the objective function contribution in (19). The variable $u_{L,j}^{P,k}$ is a state variable, as it is coupled in time in

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(16). As will be shown in Section 3.3, this means that $u_{L,j}^{P,k}$ must enter the future cost function through the cuts.

2.5 Exogenous price model

The model allows for one of the exogenous price inputs to be treated as a stochastic variable. This price can be used to describe several market options in the systems at different locations. The stochastic price input is assumed given by a number of price scenarios that describe different futures. Each scenario contains one given price for each model sub-period. Each future scenario is assumed to have equal probability and it is also assumed that there is at least one scenario corresponding to each future inflow and wind energy scenario. In the forward simulation of the SDDP algorithm these price scenarios are used directly. In the backward simulation of the SDDP algorithm a Markov chain model is used to describe the stochastic price. The Markov chain parameters are estimated based on the price scenario input. A detailed description of the SDDP algorithm with exogenous stochastic price is found in [13] and the Markov chain estimation procedure is described in [26].

3 Solution method

The overall optimisation problem is solved by a combination of SDDP and SDP, using an approach which is very close to that described in [13]. This section first provides a general introduction to the SDDP algorithm, and then emphasises on the amendments that are necessary for the present model.

3.1 Problem decomposition

We recall that by using dynamic programming principles and representing the future expected cost functions by hyperplanes or cuts, the problem is broken down to solving a sub-problem for each week and with given values of inflows, wind energy and exogenous price. The sub-problem objective function was formulated in (2) subject to the constraints (3)–(8) and (13)–(18). In Section 3.3 it is described how the cuts in (6) can be amended to include the state variables for start-up costs.

The algorithm builds an operating strategy (represented by cuts) iteratively. The two basic steps of a main iteration are illustrated in Fig. 1 and discussed below. Note that the price dimension was omitted for simplicity of the illustration.

3.1.1 Forward simulation: From the initial state $\boldsymbol{V}^0$, $\boldsymbol{z}^0$ and $u_L^{P,0}$ the system is simulated for a set of scenarios comprising realisations of stochastic variables. For a given scenario, the sub-problem described by (2)–(8) and (13)–(18) is solved for each week. That is, we assume that all stochastic variables for a given week are known at the beginning of that week. Subsequently, the simulated state at the end of the week is passed forward as an initial state for the subsequent week. The forward simulation gives an updated set of state trajectories and a (sample) expected cost. This procedure is illustrated by following the black (thicker) lines in Fig. 1 forward in time for each scenario $\{s_1, s_2, s_3\}$.

3.1.2 Dynamic backward recursion: Cuts at the end of week T can be obtained from the final value function ϕ. For each state trajectory from the forward simulation one starts from the state at the end of week T–1, and for each

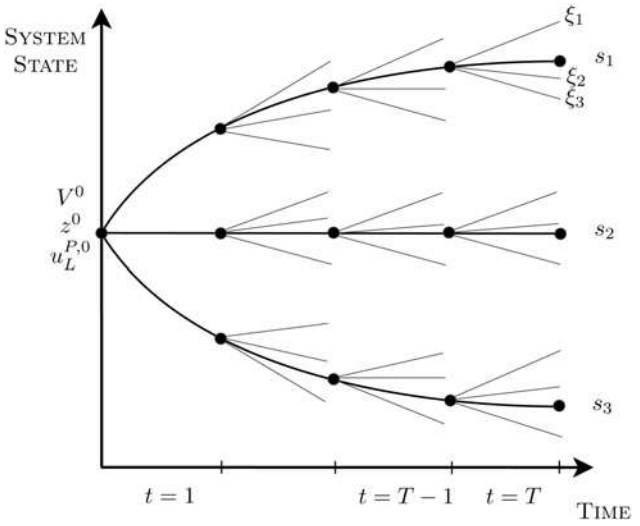


Fig. 1 *Illustration of a main iteration in the SDDP algorithm*

Each branch corresponds to a realisation of stochastic variables, and each filled circle indicates a system state. Thick lines are trajectories obtained in the forward simulation. Thin lines denote operations as decided for each stochastic outcome in the backward recursion.

realisation of stochastic variables one computes the optimal operation for week T. This is illustrated for realisations $\{\xi_1, \xi_2, \xi_3\}$ in Fig. 1. From the sensitivities of the objective function to the initial state values, new cuts at the end of week T–1 are obtained. These cuts are constructed by averaging contributions over realisations of stochastic variables for a particular state. Thus, realisations $\{\xi_1, \xi_2, \xi_3\}$ in week T for scenario s_1 in Fig. 1 will be used to construct one cut. If the lag-one model (4) is perfectly valid for the normalised inflow, then ξ_t should be independent of z_{t-1}, and thus ξ_t is uncorrelated from stage to stage. Therefore the cuts constructed will be valid for all scenarios $\{s_1, s_2, s_3\}$ at time T–1 in Fig. 1 [27]. Then one carries out the same procedure for week T–1 and so on. This gives an updated operating strategy and a lower bound for the expected cost.

Convergence is declared when the forward cost sample estimate and the lower bound from the backward recursion come close enough. This may be a slow process, or cost gaps exist, so in practice one often checks the change in trajectories from one iteration to the next, stopping after a fixed maximum number of main iterations.

If exogenous price data are used, the price dimension is treated in an SDP manner, as described in [13], and then the procedure for each week in the backward recursion is repeated N_P times, starting at each of the given discrete price states.

We assume that a set of initial trajectories is always available, so that the algorithm can start at the backward recursion step. Such a set of trajectories may for instance be obtained as percentage reservoir contents from aggregated models.

3.2 Handling of wind power stochasticity

In the forward simulation part of the SDDP algorithm, we let historical records of inflow, exogenous prices and wind energy define scenarios. These scenarios are generated based on the assumption that from the starting week each historical year have equal probability of occurrence in the

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future. These scenarios are updated according to the current system state (e.g. snow storage).

Values from these records are aggregated or disaggregated to fit the sub-period time resolution of the model. An alternative method would be to generate scenarios for forward simulation from a stochastic model. The use of 'observed' scenarios in the forward simulation may conserve correlations in the inflow, wind and price processes not taken into account in the stochastic models.

In the backward recursion, stochastic models for inflow and exogenous prices are used as discussed in Sections 2.1 and 2.5, respectively. One could in principle fit wind power into the same stochastic model as used for inflow, and thus capture some of the correlations between inflow and wind. This has not been tested here.

Some experiments were conducted using a simple stochastic wind-energy model in the backward recursion, as outlined below. For each historical record, the weekly sum of wind energy is calculated. The historical records are ordered according to sum energy, and then grouped into N_W groups, each with a predefined probability. For each group, new weekly values are constructed by averaging contributions from the corresponding records for each sub-period. In this way, a reduced set of wind-energy outcomes is constructed. Together with the N_V different inflow values and N_P discrete price states this gives $N_V \times N_P \times N_W$ different stochastic outcomes for each week. Consequently, the introduction of wind power stochasticity in the backward recursion will significantly increase computation time.

The outlined approach is capable of capturing the impact of weeks with extreme average wind conditions on the operation strategy. Such extremes would otherwise be averaged out. It will also operate with a correct expectation for the week. However, it relies on three assumptions that can be debated. First, the wind energy is considered independent of the other stochastic variables. Second, it ignores possible autocorrelation between weeks, that is, the weekly wind energy in one week is assumed independent of previous weeks. Thus, the generated wind-energy outcomes for one week are common for all states within that week in the backward recursion. Finally, and perhaps most important, the chosen stochastic time resolution of one week implies that the values of the stochastic variables are known for a whole week in advance. Given the volatile nature of wind conditions, a week-long stochastic time resolution is not fine enough to capture the actual need for balancing wind power.

Initial experiments with this technique indicate that wind power stochasticity tend to have little impact on the strategy in hydropower systems with large storage capacity and a modest share of wind power. For this reason, the presented case study results are based on average wind-power values in the backward recursion.

3.3 Handling of start-up state variables

Consider a system where N_S pumps or thermal generators are modelled with start-up costs. Owing to the time coupling in (16), the minimum pumping in the vector $\boldsymbol{u}_{L,t}^{P,k} = \left[u_{L,t,1}^{P,k}, u_{L,t,2}^{P,k}, \ldots, u_{L,t,N_S}^{P,k}\right]$ must be considered state variables and should enter the future cost function at the end of the week, where $k = K$. This function is represented by cuts. A sample cut from a sub-problem in week t can be written as

$$\alpha_t + \boldsymbol{\kappa}^{\mathrm{T}}\left(\boldsymbol{V}_t - \boldsymbol{V}_t^*\right) + \boldsymbol{\mu}^{\mathrm{T}}\left(\boldsymbol{z}_t - \boldsymbol{z}_t^*\right) \geq \alpha_t^* \tag{20}$$

Here V_t and z_t are state variables and the corresponding starred variables are the given initial values. Consider week $t+1$ in the backward run of the SDDP algorithm. For the first sub-period, (16) can be formulated as in (21) with a corresponding dual variable $\gamma_{t,\,j}$.

$$u_{L,t+1,j}^{P,1} - \delta_{t+1,j}^{1} \leq u_{L,t+1,j}^{P,*} \mid \gamma_{t,j} \tag{21}$$

At the beginning of the week $\boldsymbol{u}_{L,t+1}^{P,*} = \boldsymbol{u}_{L,t}^{P,K}$ is considered given as a part of the state along the current trajectory. Let (20) represent a cut at the beginning of week $t+1$, built from the solutions of the sub-problems for week $t+1$ that start from the given starred values at the beginning of week $t+1$. If the cost function at the beginning of the week is α_t, then

$$\frac{\partial \alpha_t}{\partial \boldsymbol{u}_{L,t+1}^{P,*}} = \boldsymbol{\gamma}_t \tag{22}$$

where $\boldsymbol{\gamma}_t = [\gamma_{t,1}, \gamma_{t,2}, \ldots, \gamma_{t,N_S}]$. The new sample cut at the end of week t becomes

$$\alpha_t + \boldsymbol{\kappa}_t^{\mathrm{T}}\left(\boldsymbol{V}_t - \boldsymbol{V}_t^*\right) + \boldsymbol{\mu}_t^{\mathrm{T}}\left(\boldsymbol{z}_t - \boldsymbol{z}_t^*\right) + \boldsymbol{\gamma}_t^{\mathrm{T}}\left(\boldsymbol{u}_{L,t}^{P} - \boldsymbol{u}_{L,t}^{P,*}\right) \geq \alpha_t^* \tag{23}$$

The cut is constructed by averaging the contributions of sample cuts of type (23). After reordering the contribution can be formulated as

$$\alpha + \boldsymbol{\kappa}_t^{\mathrm{T}}\boldsymbol{V}_t + \boldsymbol{\mu}_t^{\mathrm{T}}\boldsymbol{z}_t + \boldsymbol{\gamma}_t^{\mathrm{T}}\boldsymbol{u}_{L,t}^{P} \geq b_t \tag{24}$$

where the constant b_t is given by

$$b_t = \alpha_t^* + \boldsymbol{\kappa}_t^{\mathrm{T}}\boldsymbol{V}_t^* + \boldsymbol{\mu}_t^{\mathrm{T}}\boldsymbol{z}_t^* + \boldsymbol{\gamma}_t^{\mathrm{T}}\boldsymbol{u}_{L,t}^{P*}$$

3.4 Handling of cuts and power flow constraints

Both the cuts and the power flow constraints are handled by relaxation in order to reduce computation time, as suggested in [7, 14]. That is, the problem is first solved without these constraints, to see if they are fulfilled anyway. If not, the violated constraints are added to the linear programming (LP) problem and then it is re-solved. In the current logic, iterations are first carried out on the relaxed cuts. Subsequently, when no more cuts are violated, iterations are carried out on the power flow constraints. Adding power flow constraints will change the optimal solution, which in turn may violate new cuts; thus, iterations between the cut and power flow constraint relaxation-loops are needed. Optimising the sequence and logic of these two relaxation loops to minimise computational effort may be a topic for further research.

3.5 Computing nodal prices

Once a sub-problem is solved in the final forward simulation, a vector $\boldsymbol{\pi}^k$ of nodal prices for a given sub-period k can be computed as

$$\boldsymbol{\pi}^k = \boldsymbol{\pi}_r^k \boldsymbol{e} + \boldsymbol{\Theta}\boldsymbol{\pi}_\ell^k \tag{25}$$

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doi: 10.1049/iet-gtd.2012.0639

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where π_r^k is the dual value of the power balance (10), $e = [1, \ldots, 1]^T$ and π_ℓ^k is a vector of dual values from the power flow constraints (13). π_ℓ^k contains zeros for non-binding power flow constraints.

Nodal price contributions can be split in two terms, as shown in (25). π_r^k represents the energy component that is equal for all nodes and equals the power price at the reference busbar. The second term on the right-hand side in (25) represents the congestion components.

4 Case study

The presented model was tested on data provided by Landsvirkjun, which represents the current Icelandic power system with the addition of some potential future expansions. The considered expansions comprise an high-voltage direct current (HVDC)-light cable to the UK, a pumped-storage plant and a wind power park. Transmission grid data representing today's system was provided by the icelandic TSO Landsnet. As a large share of Iceland's power production is from hydropower, the system is in principle well suited for balancing wind power and exchange with the UK market. However, as the current transmission system is weak the hydropower's ability to regulate cannot be fully utilised, as will be studied in more detail below. Although important, possible expansion of the transmission grid to better facilitate the suggested increase in production capacity is not considered in this case study.

4.1 System description

Iceland's total electricity production in 2008 was 16.5 TWh, where approximately 70% came from hydropower and 30% from geothermal power. Geothermal power plants are used to cover base load and have strong transmission capacity to the loads. Thus, their production can be subtracted from the loads in the simulations. Landsvirkjun owns and operates almost all hydropower in Iceland with approximately 1750 MW installed capacity in 11 stations. Landsvirkjun's consumers can be split in two categories. Power intensive industries have a share of 87% at a constant profile, whereas the remaining share goes to the wholesale market.

Fig. 2 illustrates the system under investigation. All transmission system lines and transformers in the figure are assigned capacity limits. Hydropower plants are indicated by the letter H, a wind power park by W, and demand by an arrow. Geothermal plants are not shown in Fig. 2 as these are not explicitly modelled. Thus, start-up costs for thermal units are not used in this case study. The future HVDC-light cable under investigation is connected to busbar b_4 providing a link to the UK power system, with a capacity of 350 MW. The UK market was modelled as an exogenous price series, with one price scenario for each corresponding record of inflow and wind power.

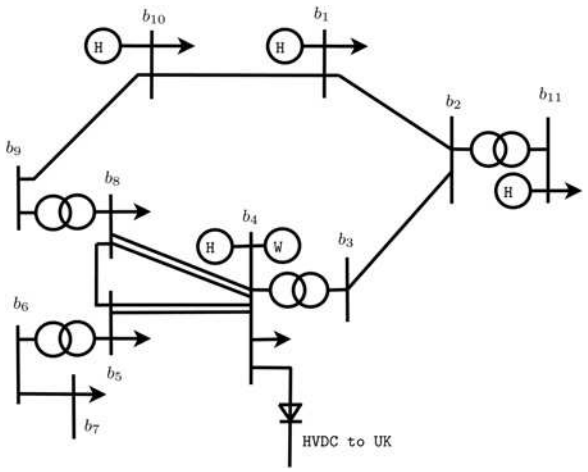


Fig. 2 *Illustration of the Icelandic power system, indicating the electrical topology and the location of loads, wind power and hydropower*

Table 1 states the number of power plants, the busbar associations, the total installed capacities and the total reservoir volumes for the five modelled watercourses. In order to add flexibility to the production system, it was extended by two projects, both connected to busbar b_4

- A pumped-storage facility at Vörðufell with 100 MW installed capacity and 20 Mm^3 upstream and downstream storage;
- A wind power park with installed capacity of 100 MW.

4.2 Simulation results

The system simulations were started in week 40 (full reservoirs) of the first year and completed at the end of the fourth year, that is a simulation period of 169 weeks. We used 54 historical weather scenarios. Available inflow records have daily time resolution and the wind power record has hourly time resolution. Simulations were carried out both for aggregate and sequential sub-periods, comprising 7 and 24 sub-periods within the week, respectively. Table 2 distinguishes the simulated cases in the arrangement of sub-periods and whether power flow constraints are included.

The single largest reservoir in Iceland, Hálslón, is located in the Jokla watercourse and is associated with busbar b_{11} in Fig. 2. The transmission network capacity between the Hálslón hydropower station and the rest of the island is only 10% of the total load at b_{11}. The majority of this load is associated with aluminium smelters. For safety of industrial operation, emptying of Hálslón should be

Table 1 Aggregate characteristics of the hydropower system, per watercourse

Name	No. plants	Busbar	Installed Cap., MW	Reservoir Vol., Mm^3
Laxa	1	b_1	18	0
Thjorsa, Sog	7	b_4	1068	2028
Blanda	1	b_{10}	150	420
Kárahnjúkar	2	b_{11}	677	2148
Vörðufell	1	b_4	100	20
Total	12		2013	4616

Table 2 Simulated cases

Case	Arrangement of sub-periods	Power flow constraints	Pumped-storage start-up cost
A	aggregate	–	–
B	aggregate	√	–
C	sequential	–	–
D	sequential	√	–
E	sequential	√	√

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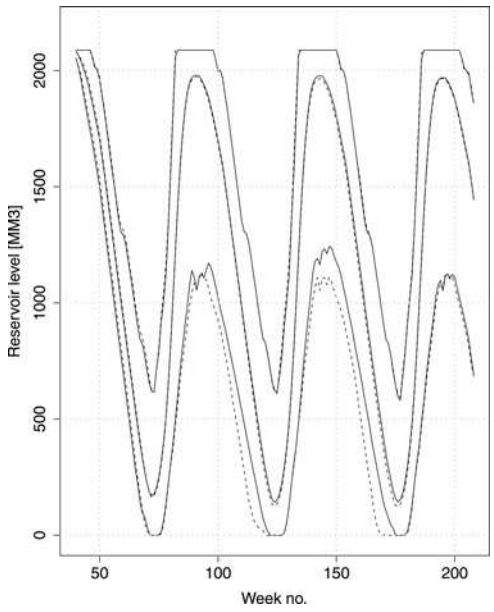


Fig. 3 *Reservoir levels in Hálslón for cases A (dashed) and B (solid). Average values and 0 and 100 percentiles*

avoided. Fig. 3 compares Hálslón's simulated reservoir levels with and without transmission constraints (cases A and B), shown as solid-drawn and dashed lines, respectively. The curves represent the average value together with 0 and 100 percentiles in Mm^3. In case B where power flow constraints are included, b_{11} will to a lesser extent see the export option at b_4. As a result, the model proposes an optimal schedule where the Hálslón reservoir level is generally higher and the risk of emptying the reservoir in extreme cases is lower than in case A.

Busbar b_4 receives power from all generation units located in the Thjorsa watercourse. This watercourse comprises several cascaded reservoirs with sizes ranging from seasonal to weekly storage. Power plants in Thjorsa are directly exposed to UK price levels through the HVDC cable connected to busbar b_4. Fig. 4 shows the simulated production as a solid-drawn line for a given week in case D for the Sigalda power plant. This power plant has an installed capacity of 160 MW and has significant upstream storage capacity. The simulated nodal prices at b_4 (in a scaled monetary unit) are shown as a dashed line in Fig. 4. The scheduled production in Sigalda fluctuates according to the UK prices; it is regulated up during day-time peaks and down at night-time and on weekends. The simulated nodal prices shown in Fig. 4 correspond to the UK prices, except from the peaks where the nodal prices at b_4 are lower. This can be explained by the limited HVDC exchange capacity. Utilisation of the HVDC cable in this particular week fluctuates between full export of 350 MW in UK price-peak periods and import at low UK prices. Note that there are no HVDC ramping constraints in the model.

Fig. 5 shows the simulated utilisation of the HVDC cable for all cases. As expected, the cable is primarily used for exporting power (positive values) to the UK. In case A, which is the least constrained case, maximum export takes place 67% of the time. Adding power flow constraints in case B, reduces this number to 62%. This result is related to the reservoir levels presented in Fig. 3; the Hálslón reservoir is used extensively to generate power for export in case A. Fig. 5 highlights the importance of adding realistic system constraints when balancing a hydropower system against external and price volatile markets and wind power. Using both a fine time resolution and power flow constraints in cases D and E clearly shows that operation of the cable is limited by the transmission system. Cases A–C clearly overestimates the use of the cable for both export and import, and thus the total profitability of the cable.

In case E we introduced a cost per start-up and a minimum generation of 30 MW for the pumped-storage plant in Vörðufell. The duration curves for pumping power for cases D and E are shown in Fig. 6. As expected, the pump utilisation decreases when introducing a start-up cost. Furthermore, Fig. 6 shows that the pump is allowed to operate a significant amount of time below the minimum capacity. This indicates that the approximate modelling is not perfect, but can be further explained by looking at the frequency of entering pumping mode for the two cases. In case D, the plant shifts from generating in a previous sub-period to pumping in 6.54% of the simulated

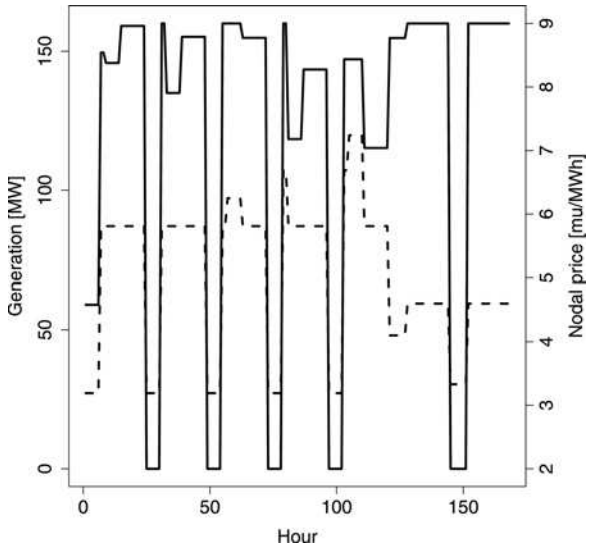


Fig. 4 *Simulated power production from the Sigalda power plant (solid) and nodal prices (dashed) at busbar b_4 for a given week in case D*

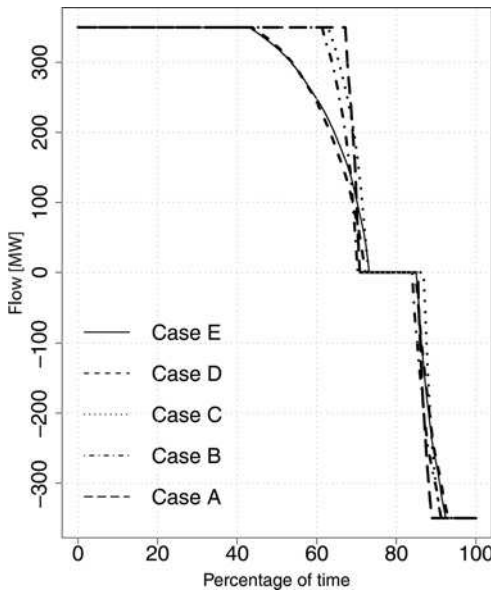


Fig. 5 *Load duration curves for the HVDC cable for cases A–E*

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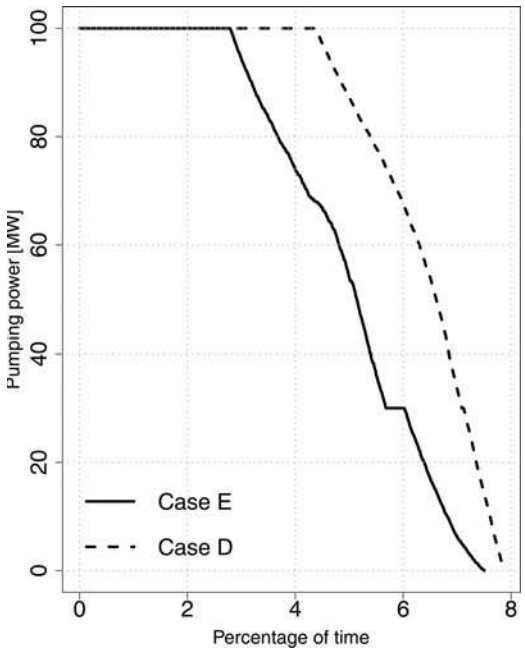


Fig. 6 *Duration curves for pumping power in Vörðufell for cases D and E*

sub-periods, whereas this occurs in only 0.12% of the sub-periods in case E. Thus, having made a pump start-up decision (either full or partial), the benefit of staying in pumping mode is recognised by the model in case E. The introduction of start-up cost also impacts prices at busbar b_4; the mean and standard deviation of the nodal price increases by 0.34% and 11.12%, respectively, compared with case D.

4.3 Computational performance

All simulations were carried out using parallel processing on a multi-core shared-memory computer with 128 GB RAM. The CPLEX LP Solver (version 11) was used for solving the sub-problems. The model was run with 12 discrete inflow noise levels and seven discrete price states in the backward recursion. To limit the computational effort the simulations referred to in Section 4.2 used the mean wind power values in the backward recursion. A maximum of 60 main iterations in the SDDP-algorithm was allowed.

As there were 54 weather scenarios to be evaluated for 169 weeks, a total of $54 \times 169 = 9126$ sub-problems were solved in each forward run for all cases. In each backward run this number increases with a factor equal to the number of stochastic outcomes, that is, $12 \times 7 = 84$.

The modelling of sequential sub-periods significantly increased computation time in the presented example. Cases A and E were both run on 12 cores, using approximately 5 and 31 h, respectively. If one does not consider the relaxation loops, the same number of LP problems are solved in the two cases. Thus, it is primarily the increased size of a single LP problem that contributes to the increase in computation time.

5 Conclusion

An SDDP-based model was presented, suitable for solving the medium- and long-term hydro thermal scheduling problem considering inflow to hydropower reservoirs, wind power and exogenously given prices as stochastic variables. The incorporation of approximate start-up costs for pumps and generating units is a novel modelling feature, where the operating states of these units enter the cuts describing the future cost functions in the SDDP approach. Furthermore, the model incorporates the treatment of detailed wind power in the forward simulation of the SDDP algorithm, and approaches for handling wind power stochasticity in the backward recursion are proposed.

The model was tested in a case study of the Icelandic power system. In general, the results indicated that when using a finer time resolution, and thus adding more constraints to the system, the simulations became more realistic. This finding was illustrated and discussed for both aggregate and detailed types of results. In particular, modelling of the limited transmission capacity between the major watercourses had a significant impact on the estimated balancing potential on Iceland. Simulations without power flow constraints clearly overestimated the capability of exporting power to the UK market and gave unrealistic reservoir operation. Finally, it was demonstrated that the utilisation of a pumped-storage plant was significantly impacted by introducing a start-up cost on pumping. The case study illustrates how this modelling feature can be used to obtain more realistic plant schedules and power prices.

Clearly, the improved system representation significantly increases computation time, which stresses the importance of parallel processing to obtain acceptable simulation times.

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